

Experimental Evaluation and Hardware Overhead Analysis of Zero-Blocking Optimization in a 4x4 Tensor Grid Accelerator

Objective

To quantitatively evaluate the operation of the Zero-Blocking circuit (zero detection and gating) in a 4x4 matrix multiplication accelerator by measuring the circuit's activity (toggling activity) as a function of input data sparsity.

Methodology & Experimental Setup

Simulation Tools: Icarus Verilog for compilation, and GTKWave / Shell scripting (grep) for extracting activity metrics from VCD files.

Evaluation Metric: The number of switches (toggling activity) recorded in the VCD file during a fixed-duration simulation was used as the direct indicator of **dynamic power consumption**.

Test Scenarios: A fixed, dense Matrix B and 5 different sparsity levels for Matrix A (ranging from 0 to 4 disabled/zeroed-out Tensor Cores).

Note: A different testbench module was written to simulate each matrix multiplication. This module is not included in this repository, as it is already demonstrated in `tb.v`. However, a terminal snippet is provided to showcase the process.

```
VCD info: dumpfile zero_blocking_tb.vcd opened for output.
Matrix Multiplication Result:
-----
A x B = C
A = [ 1  2  2  1 ]
    [ 1  1  2  2 ]
    [ 1  2  2  1 ]
    [ 1  1  1  2 ]
-----
B = [ 2  1  1  2 ]
    [ 2  1  1  2 ]
    [ 2  1  1  2 ]
    [ 2  1  1  2 ]
-----
C = [ 12  6  6  12 ]
    [ 12  6  6  12 ]
    [ 12  6  6  12 ]
    [ 10  5  5  10 ]
zero_blocking_tb.v:177: $finish called at 75 (1s)
➤ grep -c '^[0-9a-zA-Z!]\|^\#' zero_blocking_tb.vcd
1951
```

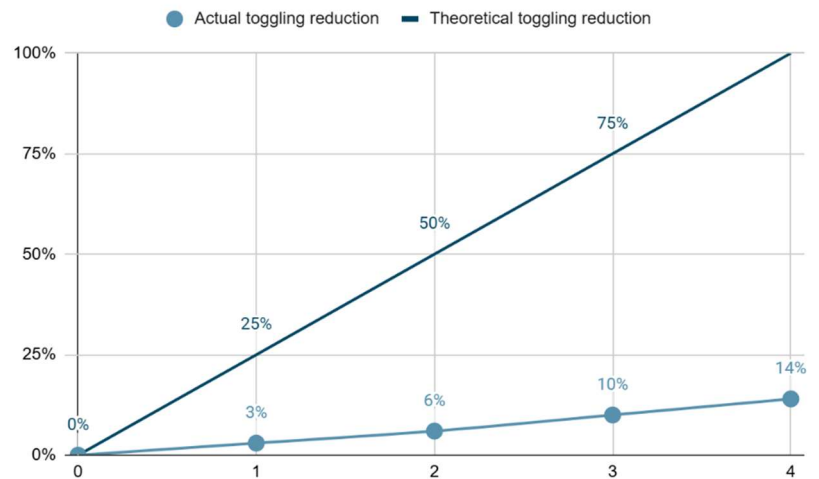
Important Context: **All documented performance benchmarks and measurements were captured with the zero-blocking optimization module fully activated.** When the zero-blocking feature is disabled under identical dense workloads, the circuit produces an equivalent baseline level of switching activity, regardless of matrix sparsity.

Analysis & Observations

<i>Disabled Tensor Cores</i>	<i>Detected switches</i>	<i>Switching reduction percentage</i>
0	1951	0%
1	1893	3%
2	1835	6%
3	1753	10%
4	1671	14%

The results show a steady, nearly linear decrease in activity as sparsity increases, with each additional zero-block removing approximately 70 switches on average.

Since each tensor core accounts for 1/4 of the grid, theoretically for each disabled tensor core a 25% switching reduction would be expected. **However experimental data show an average of 3.5% switching reduction per disabled tensor core, with a maximum reduction of 14%** (for a zeroed-out matrix).



This discrepancy is attributed to the practical hardware cost (logic/control overhead) of the detector and multiplexers, which consumes a portion of the gains at small scales (4x4).

Conclusions

Zero-Blocking technology proves to be functional and efficient, offering scalable activity savings of up to 14% in sparse arrays.

It represents a scale-dependent architectural solution: ideal for larger arrays (e.g., 16x16) and extremely sparse data, where the benefit outweighs the fixed overhead of the control logic.